most but not all string theorists was that eleven-dimensional supergravity was a mathematical oddity without any connection to the physics of string theory.12

Glimmers of M-Theory

The view now is very different. At Strings ‘95, Witten argued that if we start with the Type IIA string and increase its coupling constant from a value much less than 1 to a value much greater than 1, the physics we are still able to analyze (essentially that of the BPS saturated configurations) has a low-energy approximation that is eleven-dimensional supergravity.

When Witten announced this discovery, it stunned the audience and it has since rocked the string theory community. For almost everyone in the field, it was a completely unexpected development. Your first reaction to this result may echo that of most experts in the field: How can a theory specific to eleven dimensions be relevant to a different theory in ten?

The answer is of deep significance. To understand it, we must describe Witten’s result more precisely. Actually, it’s easier first to illustrate a closely related result discovered later by Witten and a postdoctoral fellow at Princeton University, Petr Hořava, that focuses on the Heterotic-E string. They found that the strongly coupled Heterotic-E string also has an eleven-dimensional description, and Figure 12.7 shows why. In the leftmost part of the figure, we take the Heterotic-E string coupling constant to be much smaller than 1. This is the realm that we have been describing in previous chapters and that string theorists have studied for well over a decade. As we move to the right in Figure 12.7, we sequentially increase the size of the coupling constant. Prior to 1995, string theorists knew that this would make the loop processes (see Figure 12.6) increasingly important and, as the coupling constant got larger, would ultimately invalidate the whole perturbative framework. But what no one suspected is that as the coupling constant is made larger, a new dimension becomes visible! This is the “vertical” dimension shown in Figure 12.7. Bear in mind that in this figure the two-dimensional grid with which we begin represents all nine spatial dimensions of the Heterotic-E string. Thus, the new, vertical dimension represents a tenth spatial dimension,